

Q1: Multiple choice questions

[7 marks]

1. Which statement best describes the role of a key in encryption?
 - a) The key determines which encryption function is used, making decryption difficult if the function is unknown.
 - b) The key contains a secret encryption function along with its parameters. A password is required to activate the key. ✓
 - c) All encryption functions are public, only the key is secret. It contains the parameters used for both encryption and decryption.
 - d) The key allows users to avoid reinstalling encryption software when the technology or encryption functions change.
2. In the RSA algorithm, what is the value of $\phi(n)$ if the two prime numbers chosen are $p=13$ and $q=17$?

ϕ

 - a) 292
 - b) 192 ✓
 - c) 221
 - d) 30
3. What is the effective key size of 3DES with three keys?
 - a) 56 bits
 - b) 112 bits
 - c) 128 bits
 - d) 168 bits ✓
4. What is the primary purpose of using a recipient's public key in asymmetric cryptography?
 - a) To create a digital signature for non-repudiation x
 - b) To decrypt a message that was encrypted with the recipient's private key
 - c) To encrypt a message to ensure confidentiality for the recipient ✓
 - d) To verify the integrity of a received message x
5. What is a critical vulnerability in the basic Diffie-Hellman key exchange protocol?
 - a) It uses weak prime numbers
 - b) The shared secret is too short
 - c) It is vulnerable to a Man-in-the-Middle (MITM) attack ✓
 - d) It cannot be used with symmetric ciphers
6. Which of the following algorithms use asymmetric encryption?
 - a) 3DES (TripleDES)
 - b) RSA ✓
 - c) Monoalphabetic ciphers
 - d) AES

Q2: Short answer questions

3

(4 + 3 + 3 marks)

A. Which information security property is being most directly impacted in the following scenarios?

Choose one from Confidentiality, Integrity, Availability, Authenticity, Accountability	Accountability
A server crash deletes all customer activity logs, making it impossible for the system administrator to determine which user was responsible for a specific action.	Availability
A programmer introduces a malicious backdoor into a software update that only they know about, allowing them to gain control of customer systems later. Which security property is violated when the hacker later accesses the systems?	Confidentiality, Authenticity
A university stores sensitive research data without encryption. An attacker gains unauthorized access and downloads the files. Which two security principles are most clearly violated?	Integrity
During an attack, a hacker modifies transaction records so that amounts are changed but logs show no trace of modification.	

B. A financial technology firm operates a distributed trading platform that manages high-frequency stock transactions. The system comprises three main components: a Pricing Engine (calculates real-time values), a Transaction Logger (writes all trades to an unalterable audit trail), and a User Authentication Module (handles user login/session state). A key challenge is that all three components currently rely on the same shared library for basic logging, time-stamping, and cryptographic hashing. Furthermore, the code for the Pricing Engine is overly complex, containing hundreds of functions that manage both calculation logic and database interactions. Suggest two most important design principles that you could apply to improve the system's security.

- Least Common Mechanism
- Economy of Mechanism

1.5

C. Alice wants to send a confidential contract to Bob. She also needs to prove that she, and not someone else, sent it. Bob needs to be sure the contract hasn't been altered. What should Alice do?

She can use digital signature, when she is ready to send. Her plaintext she should use her private key to sign her message when Bob receives it, he can use Alice's public key to verify it. Ensure if the message was sent. & signed by Alice in fact.

Q3: Practical problems

A. Encrypt the message

If we have war with the United States, we will have no hope of winning using the following transposition technique. Fill in the grid below, row by row from left to right. Do not place a letter in a cell containing a *.

i. Encrypt the message by reading down column one, then down column two, etc.

(4)

10	2	8	1	5	3	7	4	6	9
I	F	W	*	E	H	A	V	E	W
A	*	*	R	W	I	T	H	*	T
H	E	V	N	I	*	*	*	T	E
D	S	T	A	*	T	E	S	*	W

Ciphertext = *RNA F*ES HI*T VH*S EWI* E*T* AT*E W*VT.

ii. What problem/s would we face if we tried to cryptanalyze a message encrypted with this method?

WTEN
I A I H D

The ~~first~~ plaintext can be determined as with. frequently content analysis on the frequency/definition of words stays the same, hence it can easily be decrypted.

B. The following message is encrypted using the Vigenère cipher.

UQCG OQCG UOSV UQCG

What possible length could the keyword have and why? Show proper working.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
UQCG OQCG UOSV UQCG

(4)

Possible repetition.
words

1st Appearance.

2nd Appearance.

Difference

gcd.

UQ	1	9	8	1, 2, 4, 8
UQCG	1	13	12	1, 2, 3, 4, 6, 12
CG	3	7	4	1, 2, 4
UQ	9	13	4	1, 2, 6
CG	7	15	8	1, 2, 4, 8

Appears again.
Appears again.

keyword length = 4, because when

he words repeated most often matched 4 as the gcd, and the keyword length of repetition is also 4. UQCG suggesting that the keyword is 4.

①

101010

row = (10)₂
column = (0101)₂

c. The DES S-box-1 table is given below. If it gets an input of (101010), what is the corresponding output? Output = 06.

row = 2

column = 5

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	08
2	04	01	12	08	13	00	02	11	15	12	09	07	03	10	05
3	15	12	08	02	04	09	01	07	05	11	03	14	10	06	13

D. Given the following state matrix (in hex) for an AES decryption round, perform the InvShiftRows followed by InvSubBytes operation. The AES InvSubBytes substitution table is provided below.

63 C9 FE 30
26 F2 F2 63
7D D4 C9 C9
63 82 D4 FA

63 C9 FE 30 → shift 0 byte to right
F2 → F2 → 63 → 26 → shift 1 byte to right
C9 → C9 → 7D → D4 → shift 2 byte to right
FA → 63 → 82 → D4 → shift 3 byte to right

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	52	09	6A	D5	30	39	A5	38	BF	40	A3	9E	81	F3	D7
1	7C	E3	39	82	9B	2E	1E	87	24	8E	43	44	C4	D1	19
2	84	7E	94	32	A6	C7	23	3D	EE	4C	98	0B	42	1A	C3
3	08	2E	A1	66	28	D9	24	B2	76	5B	A2	49	6D	8B	D1
4	72	F8	E6	64	86	98	16	D4	A4	SC	CC	5D	65	B6	92
5	6C	70	48	50	FD	E0	DA	5E	15	46	57	A7	8D	9D	84
6	90	DB	8C	00	8C	BC	0A	DA	F7	F4	59	05	BS	B3	45
7	DB	3C	1F	8F	CA	4F	0E	02	C1	AF	BD	03	01	13	SA
8	3A	91	11	41	4E	67	DC	EA	97	F2	CB	CF	F0	B4	E9
9	96	AC	74	22	E7	AD	35	85	E2	F9	37	E8	1C	75	DF
A	47	F1	1A	71	1D	29	C5	89	6F	B7	62	0E	AA	18	BE
B	1C	56	7E	4B	C6	D2	79	20	9A	D8	C0	F1	78	CD	SA
C	1F	DD	AS	33	88	07	C7	31	B1	(2)	10	S9	27	80	EC
D	60	51	7E	A9	(19)	BS	4A	UD	2D	E5	7A	9F	93	C9	9C
E	A0	F0	3B	4D	AE	2A	E5	B0	C8	1B	BB	3C	83	53	99
F	17	2B	(04)	7E	BA	77	D6	26	E1	69	(14)	63	55	21	DC

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Intermediate output after InvShiftRows:

Final matrix after InvSubBytes:

63 C9 FE 30
26 F2 F2 63
7D D4 C9 C9
63 82 D4 FA

00 12 0C 08
23 04 04 00
13 19 12 12
00 11 19 14

misis
26.

Q4: Numerical questions

[2 + 2 + 3 + 3 marks]

Alice and Bob want to set up an RSA public-key cryptosystem. Alice chooses two small prime numbers to generate her keys.

Given Parameters for Alice: $p = 3$ and $q = 11$
Messages to be encrypted: $M_1 = 2$ and $M_2 = 5$

Questions

- Compute the ~~modulus~~ n and Euler's totient function $\phi(n)$ for the given primes.
- Calculate d for $e = 7$.
- Describe the steps for generating the public/private key pair. You must state the conditions/properties of any values to be selected or calculated.
- Encrypt the two messages to find the ciphertexts C_1 and C_2 .

(i) $n = p \times q = 3 \times 11 = 33$

(2)

$$\phi(n) = (p-1)(q-1) \\ = (3-1)(11-1) = 2 \times 10 = 20$$

(ii) $d = \frac{\phi(n)+1}{e} = \frac{20+1}{7} = 3$

(2)

(iii) Public key = (e, n)

$= (7, 33)$

Private key = (d, n)

$= (3, 33)$

(3)

- First we calculate the $n = p \times q$
or then $d = (p-1)(q-1)$
or we should find prime no. a coprime
of e and $\phi(n)$ where condition
is $1 < e < \phi(n)$ or
from $\gcd(e, \phi(n)) = 1$
 d is a coprime of e .
- d is calculated by
 $d \cdot e \bmod \phi(n) = 1$
 $d = \frac{\phi(n)+1}{e}$
- Public key is combination of
 (e, n) or private key is
 (d, n) .

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Extra Page for Q4 or Rough Work

(iv) $C_{12} M_1^e \text{ mod. } n.$

$$2^7 \text{ mod. } 33$$

$$2^{128} \text{ mod } 33$$

$$2^{29}$$

13

$C_{22} M_2^e \text{ mod } n.$

$$2^5 \text{ mod } 33$$

$$2^{78125} \text{ mod } 33$$

$$2^{14}$$

$$\begin{array}{r} 3 \\ 33 \overline{) 128} \\ \underline{-99} \\ 29 \end{array}$$

$$\begin{array}{r} 2367 \\ 33 \overline{) 78125} \\ \underline{-66} \\ 121 \\ \underline{-99} \\ 222 \\ \underline{-198} \\ 245 \\ \underline{-231} \\ 14 \end{array}$$